

QIONG LIU, PH.D.

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Authorized to work in the U.S. without sponsorship

Professional Summary

AI Scientist and Biomedical Imaging Researcher with expertise in **cardiovascular AI, cardiac PET imaging, respiratory/cardiac motion correction, physiological signal extraction, tracer kinetic modeling, and quantitative imaging analysis**. Experienced in developing deep learning and physics-informed methods for **cardiac motion compensation, low-count denoising, parametric imaging, and clinically robust quantitative analysis** in dynamic PET. Strong background in translating cardiovascular imaging workflows into AI-driven solutions through close collaboration with cardiologists, physicians, imaging scientists, and product teams.

Core Expertise

Cardiovascular AI & Quantitative Imaging: Cardiac PET imaging, tracer kinetic modeling, parametric imaging, physiological signal analysis, ECG-correlated validation, quantitative biomarker extraction

Motion Correction & Signal Processing: Respiratory/cardiac dual gating, frequency-domain analysis, dynamic PET motion compensation, deformation field fusion

Machine Learning: Deep learning, self-supervised learning, physics-informed AI, representation learning, diffusion models, transformers

Medical Imaging: PET reconstruction, low-count denoising, dynamic PET, image quality assessment, clinical imaging validation

Programming: Python, PyTorch, NumPy, SciPy, MATLAB

Experience

Canon Medical Research USA

Apr 2025 – Present

AI Scientist / Reconstruction Scientist, Vernon Hills, IL

- Developed AI-driven motion correction and reconstruction methods for **dynamic cardiac PET imaging** under noisy and low-count acquisition conditions.
- Designed **automatic cardiac-respiratory dual gating algorithms** to extract physiological motion signals directly from dynamic PET sequences.
- Developed **EdgeFuse**, an edge-guided deformation field fusion framework for cardiac PET motion correction to improve myocardial boundary preservation and quantitative accuracy.
- Built end-to-end pipelines integrating preprocessing, model training, inference, motion correction, denoising, and downstream quantitative assessment.
- Collaborated with physicians, imaging scientists, and product teams to evaluate image quality, temporal consistency, quantitative robustness, and clinical usability.

Canon Medical Research USA

Jun 2023 – May 2024

Research Scientist Intern, Vernon Hills, IL

- Developed deep learning methods for low-count PET denoising while preserving quantitative fidelity for downstream dynamic and parametric analysis.
- Designed evaluation frameworks to assess robustness, generalization, image quality, and quantitative consistency across heterogeneous imaging datasets.
- Presented technical findings to guide algorithm design and translational imaging research.

Yale University PET Center

Aug 2020 – May 2025

Ph.D. Researcher, New Haven, CT

- Developed AI and quantitative imaging methods for **dynamic cardiac PET analysis, tracer kinetic modeling, parametric imaging, and low-count denoising**.
- Applied **Patlak analysis** and kinetics-informed methods to derive image-based cardiovascular biomarkers from noisy dynamic PET data.
- Performed quantitative cardiac PET analysis to evaluate treatment-related effects in transthyretin cardiac amyloidosis.
- Built self-supervised and deep image prior-based denoising methods that preserve kinetic modeling accuracy and downstream parametric quantification.
- Collaborated with cardiologists, nuclear medicine physicians, and clinical researchers to define clinically meaningful imaging endpoints and validate quantitative results.

United Imaging Healthcare
Student Intern, Wuhan, China

Jul 2019 – Aug 2019

- Supported development and validation of engineering systems for clinical applications.

Selected Cardiovascular AI Projects

Automatic Cardiac-Respiratory Dual Gating for Dynamic PET

- Developed an AI-driven dual gating framework for simultaneous extraction of respiratory and cardiac motion signals directly from dynamic PET data.
- Combined autoencoder feature learning, PCA, and frequency-domain analysis to separate physiological motion components.
- Validated extracted cardiac signals against ECG recordings and visually evaluated respiratory motion using high-temporal-resolution PET mini-frame movies.

EdgeFuse: Edge-Guided Motion Correction for Cardiac PET

- Developed a spatially adaptive deep learning framework for cardiac PET motion correction using edge-guided deformation field fusion.
- Combined deformation fields with different smoothness constraints to preserve myocardial boundaries while maintaining smooth background motion.
- Improved motion compensation, lesion localization, and quantitative accuracy in cardiac PET imaging.

Dynamic Cardiac PET Quantification and Kinetic Modeling

- Developed quantitative analysis pipelines for dynamic cardiac PET using Patlak-guided and kinetics-informed methods.
- Evaluated treatment-related effects in transthyretin cardiac amyloidosis using parametric imaging biomarkers.
- Improved robustness of quantitative cardiovascular imaging under noisy and low-count acquisition conditions.

Low-Count and Dynamic PET Denoising

- Developed personalized, population-based, and self-supervised denoising frameworks for dynamic and low-count PET imaging.
- Preserved quantitative fidelity and downstream kinetic modeling accuracy during denoising.
- Designed evaluation protocols emphasizing image quality, temporal consistency, and parametric quantification.

Patents

Edge-Guided Deformation Field Fusion for PET Motion Correction

Patent Pending

Lead Inventor

- Developed an AI-driven cardiac PET motion correction framework that adaptively combines deformation fields to preserve myocardial boundaries and improve quantitative accuracy.

Automatic Data-Driven Cardiac-Respiratory Gating

Patent Pending

Lead Inventor

- Developed an AI-based dual gating framework for simultaneous extraction of respiratory and cardiac physiological motion signals directly from dynamic PET data.

Education

Yale University

Aug 2020 – May 2025

Ph.D. in Biomedical Engineering

Thesis: Improving Image Quality and Quantification Accuracy for Static and Dynamic PET

Huazhong University of Science and Technology

Sep 2016 – Jun 2020

B.S. in Biomedical Engineering, Minor in Computer Science

GPA: 3.94/4.00

Selected Publications

- Liu Q., et al., *Parametric cardiac imaging with ^{18}F -flutemetamol PET to evaluate the impact of tafamidis in patients with transthyretin cardiac amyloidosis*, **Journal of Nuclear Medicine**, 2026.
- Liu Q., et al., *Patlak-guided self-supervised learning for dynamic PET denoising*, **IEEE Transactions on Radiation and Plasma Medical Sciences**, 2026.
- Liu Q., et al., *Population-based deep image prior for dynamic PET denoising: a data-driven approach to improve parametric quantification*, **Medical Image Analysis**, 2024.
- Liu Q., et al., *A personalized deep learning denoising strategy for low-count PET images*, **Physics in Medicine & Biology**, 2022.
- Liu Q., et al., *Dynamic imaging and tracer kinetic modeling of ^{18}F -flutemetamol PET for ATTR cardiac amyloidosis patients*, **SNMMI Oral Presentation**, 2023.
- Liu Q., et al., *Comparison of network structures for deep learning-based cardiac PET scatter correction*, **IEEE NSS MIC**, 2024.

Awards

Society of Nuclear Medicine and Molecular Imaging (SNMMI)

- Young Investigator Award, 1st Place, Cardiovascular Council Clinical Science, 2025.
- Poster Award, 2nd Place, Physics, Instrumentation & Data Sciences Council, 2024.
- Young Investigator Award, 2nd Place, Cardiovascular Council Clinical Science, 2023.

Mitacs

- Globalink Research Award, 2019.